ETALENESS TRANSFERS BETWEEN A DILATATION AND ITS FORMAL MODEL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof handles the open part by the isomorphism over U, writes X_{/T} as the colimit of the infinitesimal neighbourhoods, deduces etaleness of each base change from g, and applies the infinitesimal etaleness criterion

Editorial note. The label lemma-output-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the dilatations theorem (theorem-dilatations-general of the same chapter) and the correspondence between f : X' -> X and g : W -> X_{/T}
- etale f gives etale formal completion (lemma-map-completions-etale of the same chapter)
- the reduced induced closed subspace Z on T, its infinitesimal neighbourhoods Z_n, and X_{/T} = colim Z_n
- the identification of f^{-1}(Z_n) with X' x_X Z_n and etaleness of its map to Z_n from the assumed etaleness of g
- etaleness checked on infinitesimal neighbourhoods (More on Morphisms of Spaces, Lemma spaces-more-morphisms-lemma-check-smoothness-on-infinitesimal-nbhds)

Source reference. Original reference 0GI0: sources/stacks-restricted/pinned-original-source.tex, source lines 6026–6026. Prerequisite pointer invoked at proof line(s) 7345; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0APQ: sources/stacks-chow/tags.txt, source lines 13483–13483. Prerequisite pointer invoked at proof line(s) 7359; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 047X: sources/stacks-chow/tags.txt, source lines 5524–5524. Prerequisite pointer invoked at proof line(s) 7353; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0ARB: sources/stacks-restricted/pinned-original-source.tex, source lines 7015–7015. Prerequisite pointer invoked at proof line(s) 7338; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With assumptions and notation as in Theorem 0ARB let $f : X' \to X$ correspond to $g : W \to X_{/T}$. Then f is étale if and only if g is étale.*

Proof. If f is étale, then g is étale by Lemma 0GI0. Conversely, assume g is étale. Since f is an isomorphism over U we see that f is étale over U. Thus it suffices to prove that f is étale at any point of X' lying over T. Denote $Z \subset X$ the reduced closed subspace whose underlying topological space is $|Z| = T \subset |X|$, see Properties of Spaces, Definition 047X. Letting $Z_n \subset X$ be the nth infinitesimal neighbourhood we have $X_{/T} = \operatorname{colim} Z_n$. Since $X'_{/T} = W \to X_{/T}$ we conclude that $f^{-1}(Z_n) = X' \times_X Z_n \to Z_n$ is étale by the assumed étaleness of g. By More on Morphisms of Spaces, Lemma 0APQ we conclude that f is étale at points lying over T. □